

Stat 5200, Module 3: Completely Randomized Designs and ANOVA

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chapters 3-6 -

Overview

Objectives and overview

- This material is based Chapters 3–6 on the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Module outcomes include:
 - You will be able to describe a completely randomized design.
 - You will be able to describe the basic model used for ANOVA.
 - You will know how to perform an analysis of data from a CRD using ANOVA.
 - You will be familiar with contrasts and how they are used to perform inference.
 - You will understand the fundamental issues regarding multiple testing, and learn basic methods to overcome these issues.
 - You will know how to assess model assumptions.

Completely Randomized Designs

Completely Randomized Design

- Completely randomized design (CRD) is the simplest design for comparing treatments and is sometimes the preferred design.

$$i \in \{1, 2, 3, \dots, g\}$$

- **Setup:**
 - There are g treatment groups.
 - $\underline{N}$ experimental units (EU)s available.
 - You want sample sizes of $n_1, n_2, \dots, n_g$, with $\sum_{i=1}^g n_i = N$.

Completely Randomized Design II

- **Idea:** All possible sub-group composition are equally as likely.

Performing a Completely Randomized Design

- Randomly choose n_1 EU's for treatment 1.
- Randomly choose n_2 EU's for treatment 2.
- ...
- Randomly choose n_{g-1} EU's for treatment $g - 1$.
- The remaining n_g EU's are assigned to treatment g .

- On a computer, this can be done by permuting $1, 2, \dots, N$, and assigning the first n_1 numbers to treatment 1, the next n_2 to treatment 2, etc.

set.seed(...)
$$\text{Sample}(1:100) = \overbrace{74, 16, 23, \dots}^{n_1} \quad N=100$$

Advantages and Disadvantages

- Advantages
 - Simple to do; all you need is one random permutation.
 - Simple to explain and understand.
 - Simple to analyze.
 - No restrictions on the sample sizes (other designs may restrict n_i).
 - Easy with missing values.
- Disadvantages
 - Not always possible or cost-effective.
 - The accuracy of the experiment depends on variability among the EUs, not just the variability among treatment groups. Thus, it may require more EUs (\$\$\$\$).

Example: Acid Rain

Wood and Bormann (1974) studied the effect of acid rain on trees. One experiment used 240 six-week-old yellow birch seedlings. These seedlings were divided into five groups of 48 *at random*, and the seedlings within each group received an acid mist treatment 6 hours a week for 17 weeks. The five treatments differed by mist pH: 4.7, 4.0, 3.3, 3.0, and 2.3. Otherwise, the seedlings were treated identically. After the 17 weeks, the seedlings were weighed, and total plant (dry) weight was taken as response.

Questions:

- What type of design is this, and can you describe why?
- What are the EUs, and how many are there?
- What was the response?

CRD

all EUs are treated equally, except for Random assignments.

seedlings, 240

total plant dry weight.

Example: Acid Rain II

- What compromises may have been made in this case?

- only one type of plant.
- only one measurement (mass).
- only measured on seedlings.
- No control.

Like many experiments, the researchers necessarily have to narrow down an enormous question to a workable experiment using artificial acid rain under controlled conditions.

Chapter 7 : power calculations.

power $> .80$

Models and Parameters

Introduction and definitions

$$\mathcal{N}(\mu, \sigma^2), \chi^2_{df}, t_{df}$$

- All formal statistical analysis is based on probability models.
- *EX*: The number of heads in ten tosses of a fair coin would be $\text{Binomial}(10, 0.5)$. Here, the model depends on 2 parameters: $N = 10$ the number of trials, $p = 0.5$ is the probability of heads.
- In an experiment, we may posit several different models for the data, all with *unknown* parameters. One of our objectives is often to describe which model is the best description of the data, and make *inference* about parameters.

Definition: Inference

Using data to learn properties of a probability distribution. In our case, using data to estimate parameters of a model.

Introduction and definitions II

- The models we will consider for experimental design will have two basic parts:
 - A description of the expected value, or mean, of the data. Sometimes called “model for the means”.
 - A description of how data vary around the treatment means. Sometimes called ‘model for the errors’.
- For the errors, we will generally assume that they are independent for different data values, have mean zero, and all have a common variance σ^2 .
- To perform formal tests and obtain confidence intervals, we will further assume a distribution of the errors. Generally, we assume that the errors follow a **Gaussian** (Normal) distribution.

Introduction and definitions III

- For a CRD, we start with two basic models.
- Recall: $\underbrace{g}$ treatment groups, $\underbrace{N}$ EUs.

1. The Full Model (separate group means)

$y_{ij} \leftarrow$ response, i^{th} treatment group
 $i \in 1: g$

j^{th} observation in
group i ,

$j \in 1: n_i$

$$y_{ij} \stackrel{\text{iid}}{\sim} N(\mu_i, \sigma^2)$$

↓

$$y_{ij} = \mu_i + \varepsilon_{ij},$$

$$\varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Introduction and definitions IV

2. The Reduced Model (single mean model)

$$y_{ij} \stackrel{iid}{\sim} N(\mu, \sigma^2)$$

$$y_{ij} = \mu + \epsilon_{ij} \quad \epsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$$

model (2) is a special case of (1):

model 2 is model 1, if $\mu_1 = \mu_2 = \dots = \mu_g = \mu$

model 1:

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

model 2:

$$y_{ij} = \mu + \varepsilon_{ij}$$

Defining overall mean

$$y_{ij} = \mu^* + \alpha_i + \varepsilon_{ij}$$

- It's sometimes useful to express the group means μ_i as:

$$\mu_i = \mu^* + \alpha_i.$$

In this case, we call μ^* the *overall mean*, and the α_i 's are the *i*th *treatment effects*.

- This introduces an extra parameter ($g+1$ total), which we need to address. Generally, we do this by defining μ^* and require that $\alpha_i = \mu_i - \mu^*$.

Defining overall mean II

- One classic choice is to define $\underline{\mu}^*$ as the mean of the treatment means:

g groups, g means (μ_i)

$$\mu^* = \frac{1}{g} \sum_{i=1}^g \mu_i$$

$$\alpha_i = \mu_i - \mu^*$$

$$\sum_{i=1}^g \alpha_i = 0$$

(HW problems)

Defining overall mean III



- An alternative choice (that can sometimes make hand-work simpler) instead describes μ^* as the weighted average:

$$\mu^* = \sum_{i=1}^g \frac{n_i}{N} \cdot \mu_i$$

$$3.5 - 3.6$$

↓

$$d_i = \mu^* - \mu_i$$

$$\sum n_i d_i = 0$$

Recap Thu, Sep 17

CRD: g treatment groups, $i \in \{1, 2, \dots, g\}$

n_i Eu's in each group

$$\left. \begin{array}{l} \text{group (1): } n_1 \\ \text{group (2): } n_2 \\ \vdots \\ \text{group (g): } n_g \end{array} \right\} N = \sum_{i=1}^g n_i$$

$y_{ij} \leftarrow j^{\text{th}}$ replicate. from
group i .

$$i \in 1:g, j \in 1:n_i$$

Full model: $(H_1) \quad y_{ij} \stackrel{iid}{\sim} N(\mu_i, \sigma^2) \leftarrow$

same $y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$

$y_{ij} = \underbrace{\mu^*}_{\text{global mean}} + \underbrace{\alpha_i}_{\text{treatment effect (i)}} + \varepsilon_{ij}$ ~~errors~~

Reduced model (H_0)

$$y_{ij} \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

$$y_{ij} = \mu + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

observation
 j in
group i .

" μ " (new)
(mean)

"epsilon"
(error)

$$\alpha_i = 0$$

Inference : estimate parameters
of models using data.

(1) μ^* is the mean of means

$$\mu^* = \frac{1}{g} \sum_{i=1}^g \mu_i$$

(2) μ^* is weighted mean of means

$$\mu^* = \sum_{i=1}^g \frac{n_i}{N} \mu_i$$

(2) "makes more sense"

Suppose unbalanced



(if approach (1), then if

$$\mu_1 = 5$$

$$\mu_2 = 200$$

$$\uparrow \mu^* = \frac{5 + 200}{2} = \underline{\underline{102.5}}$$

approach (2)

$$\left(\frac{2}{102}\right)5 + \left(\frac{100}{102}\right)200 \approx 196.18$$

average of all data: weighted
global
average.

Defining overall mean IV

Reference Group

- Another common choice in software is to set $\mu^* = \mu_j$, for some $j \in 1 : g$. Many software just use the last mean:

$$\mu^* = \mu_g.$$

$$\mu^* = \mu_g$$

j is "control"
or global mean

$$\alpha_i = \mu^* - \mu_i$$

$$\alpha_i = \begin{cases} \mu_i - \mu_j = \mu_j - \mu_j = \underline{\underline{0}} & i=j \\ \mu_i - \mu_j & i \neq j \end{cases}$$

$$\alpha_i = \begin{cases} 0 & i=j \\ \mu_i - \mu_j & i \neq j \end{cases}$$

Model Recap

However you choose to define μ^* and α_i 's, we end up with the following models for the data:

- *The full model:*

$$y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$$

- *The reduced model:*

$$y_{ij} = \mu + \epsilon_{ij}$$

Parameter Estimation

"estimator" is a function
(a method) of estimating
a parameter from data.

θ Some parameter
↑
"theta"
 $N(\mu, \sigma^2)$
↑ ↗
 θ

estimator for θ is denoted $\hat{\theta}$

Estimating parameters

- We want to use data to estimate μ_i 's and the variance σ^2 . From these, we'll compute μ^* and α_i 's.
- We will focus on computing **unbiased** estimates of these parameters.
- Estimate for μ_i :

$$\hat{\mu}_i = \bar{y}_{i\cdot} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$$

- The estimate for μ^* is fixed, so it depends on how we defined it.

$$i \in 1, 2, 3, \dots, g$$

$$j \in 1, 2, \dots, n_i$$

$$\underline{y_{2.}} = (y_{21}, y_{22}, \dots, y_{2n_2})$$

$\underline{y_{2.}}$ = "average of all data
in treatment group 2."

$\underline{y_{..}}$ = "average of all data"

Estimating parameters II

- Unweighted average

$$\hat{\mu}^* = \frac{1}{g} \sum_{i=1}^g \hat{\mu}_i = \frac{1}{g} \sum_{i=1}^g \bar{y}_{i.}$$

$$\hat{\alpha}_j = \hat{\mu}_j - \hat{\mu}^* = \bar{y}_{j.} - \frac{1}{g} \sum_{i=1}^g \bar{y}_{i.}$$

- Weighted average

$$\hat{\mu}^* = \frac{1}{N} \sum_{i=1}^g n_i \hat{\mu}_i = \frac{1}{N} \sum_{i=1}^g n_i \bar{y}_{i.} = \frac{1}{N} \sum_{i=1}^g \sum_{j=1}^{n_i} y_{ij}$$

$$\hat{\alpha}_j = \hat{\mu}_j - \hat{\mu}^* = \bar{y}_{j.} - \bar{y}_{..}$$

$$= \frac{1}{N} \sum_{i=1}^g \sum_{j=1}^{n_i} y_{ij} = \bar{y}_{..}$$

- Reference Group

$$\hat{\mu}^* = \hat{\mu}_j = \bar{y}_{j.}$$

$$\hat{\alpha}_i = \hat{\mu}_i - \hat{\mu}^* = \bar{y}_{i.} - \bar{y}_{j.} = \begin{cases} 0, & \text{if } i=j \\ \bar{y}_{i.} - \bar{y}_{j.}, & i \neq j \end{cases}$$

Estimating parameters III

- The only parameter left to estimate is σ^2 :

$$N(\mu_i, \sigma^2)$$

$$(\mu_1, \mu_2, \dots, \mu_g, \sigma^2)$$

↓ (Sum of square errors

$$\hat{\sigma}^2 = MS_E = \frac{SS_E}{N-g} = \frac{\sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\cdot})^2}{N-g}$$

(Mean-square of Errors)

$$3.1 - 3.5$$

Estimating parameters IV

- Above, we have described **point estimates**. In some sense, these are our “best guess” as to the value of a particular parameter.
- We also want some measure of uncertainty of this estimate. The standard frequentist way of doing this is a **confidence interval**.
- Confidence intervals for α are not very useful, because it depends on how we define them. Confidence intervals for group means are typically defined as:

Comparing Models: ANOVA

Analysis of Variance

- In our analysis of a CRD, our models focus on the *mean responses* of the treatment groups.
- A basic question we might want answered: Are all the means the same, or are some different?
- We can express this same question mathematically:

Analysis of Variance II

- The basic idea for many statistical tests (including ANOVA) follows *Occam's Razor*: we use the simplest model that is consistent with the data; we will only use a more complicated model if the data indicate that the more complicated model is needed.
- How do we know if a more complex model is needed? We compare **residuals**.

Analysis of Variance III

- The lower the SSR, the more of the variance in the data is explained by the model.
- Importantly, the SSR for the reduced (single mean) model can *never* be lower than the separate model, and in practice, is nearly always larger.
- What we need now is a way to consistently decide when the separate means model is sufficiently “better” than the single means model.
- To do so, we will set up a statistical hypothesis test:

ANOVA mechanics

- ANOVA works by partitioning the total variance in the data into parts that mimic the model.
- For instance, in the separate mean model, the model says the data differ from the grand mean by a random error and a treatment effect:

ANOVA will have a similar decomposition: one part dealing with group means, one part with deviations from those means:

- Recall from Eq. 2, we have:

$$y_{ij} - \bar{y}_{\bullet\bullet} = \hat{\alpha}_i + r_{ij},$$

ANOVA mechanics III

- Finally, we can connect this back to the residuals.
- The Sum of Squared Residuals (SSR) depended on our model:

$$SSR_{\text{single means}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2 = SS_T$$

$$SSR_{\text{separate means}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2 = SS_E$$

Calibrating Effect Size

- We have established:
 - We will compare single mean against separate mean by comparing their SSRs.
 - Comparing SSR can be done by decomposing total error into two parts: $SS_T = SS_{T_{rt}} + SS_E$, and if the separate model is “much better” than the $SS_{T_{rt}}$ will be large.
- **How large is large enough?**

Calibrating Effect Size II

- The answer also lies in the decomposition of sum of squares.

ANOVA variable recap

- Sum of Square Residuals:

$$SSR = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \hat{y}_{ij})^2$$

- Treatment Sum of Squares:

$$SS_{Trt} = \sum_{i=1}^g \sum_{j=1}^{n_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 = \sum_{i=1}^g n_i \hat{\alpha}_i^2$$

- Error Sum of Squares:

$$SS_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2$$

ANOVA variable recap II

- Total Sum of Squares:

$$SS_T = SS_{T_{rt}} + S_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2$$

- F-statistic:

$$\frac{SS_{T_{rt}}/(g-1)}{SS_E/(N-g)} \sim F_{g-1, N-g}$$

Example in R

We are going to look at an example using the `PlantGrowth` dataset in R. This is a built-in dataset, so we don't need to download anything.

Example in R II

```
library(emmeans) # Package that we will use for analysis  
data(PlantGrowth)  
str(PlantGrowth)
```

```
'data.frame': 30 obs. of 2 variables:
```

```
$ weight: num 4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.5..
```

```
$ group : Factor w/ 3 levels "ctrl","trt1",...: 1 1 1..
```

```
levels(PlantGrowth$group)
```

```
[1] "ctrl" "trt1" "trt2"
```

Example in R III

```
pg_model <- aov(weight ~ group, data = PlantGrowth)
summary(pg_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	3.766	1.8832	4.846	0.0159 *
Residuals	27	10.492	0.3886		

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example in R IV

Now to get group means:

```
pg_emm <- emmeans(pg_model, ~ group)
pg_emm
```

group	emmean	SE	df	lower.CL	upper.CL
ctrl	5.03	0.197	27	4.63	5.44
trt1	4.66	0.197	27	4.26	5.07
trt2	5.53	0.197	27	5.12	5.93

Confidence level used: 0.95

Contrasts

Introduction to Contrasts

- An ANOVA can give us a good overview as to whether all the treatment groups have the same mean response.
- However, an ANOVA by itself does not tell which treatments are different or in what ways they differ.
 - One of the problems: There are many ways to define treatment effects, so confidence intervals are not immediately useful.
- One method to examine treatment effects is called a **contrast**.

Formal Definition

Definition: Contrasts

A contrast is a linear combination of treatment means, μ_i 's:

$$W(\mu_{1:g}) = \sum_i w_i \mu_i, \quad \text{with} \quad \sum_i w_i = 0$$

- The condition $\sum_i w_i = 0$ guarantees that $W(\mu_{1:g})$ does not depend on μ .

Formal Definition II

- Often the weights w_i themselves are referred to as a contrast.
- An **observed** contrast or estimated contrast is defined as:

$$W(\bar{y}_{1:g,\bullet}) = \sum_i w_i \bar{y}_{i\bullet} = \sum_i w_i \hat{\alpha}_i = W(\hat{\alpha}_{1:g})$$

Types of Common Contrasts

Pairwise Contrasts

- It is often of interest to compare all pairs of groups.

Types of Common Contrasts II

Comparison with a Control

- Let's suppose treatment 1 is a control.
- An obvious idea is to compare the mean or effect of the control with the average mean or effect of all non-controls.

Types of Common Contrasts III

Multiple Treatments (Factorial Designs)

- Suppose there are two treatment factors, A and B , each at two levels, 1 and 2.
- We'll denote the treatment group means as $\mu_{A1}, \mu_{A2}, \mu_{B1}, \mu_{B2}$, respectively.
- We might want to compare the averages of just one treatment type (A vs B or 1 vs 2).

Types of Common Contrasts IV

- What do we do with contrasts?
- If a contrast is large (in magnitude), that provides evidence that there is a difference between treatment levels that are specified by the contrast. Conversely, if it is small, there is little evidence of a difference.
- The common question arises: how large is “large enough” to declare the contrast is not zero?
- We can use the linearity of the model to answer this question!

Inference on Contrasts

- Under the null hypothesis and normality assumption, we can use Student's t with $df = N - g$ in tests or confidence intervals.
- Under the null hypothesis, $W(\mu_i) = \sum_i w_i \mu_i = 0$.
- The standardized estimated contrast follows a student-t distribution with $df = N - g$:

$$t = \frac{\sum_i w_i \bar{y}_{i\bullet}}{\sqrt{MS_E} \sqrt{\sum_i w_i^2 / n_i}}, \quad MS_E = \frac{\sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2}{N - g}$$

Inference on Contrasts II

- A $1 - \alpha$ confidence interval for a contrast $W(\mu_i)$ has the usual form:

estimate $\pm$ critical value $\times$ standard error

Contrasts in R

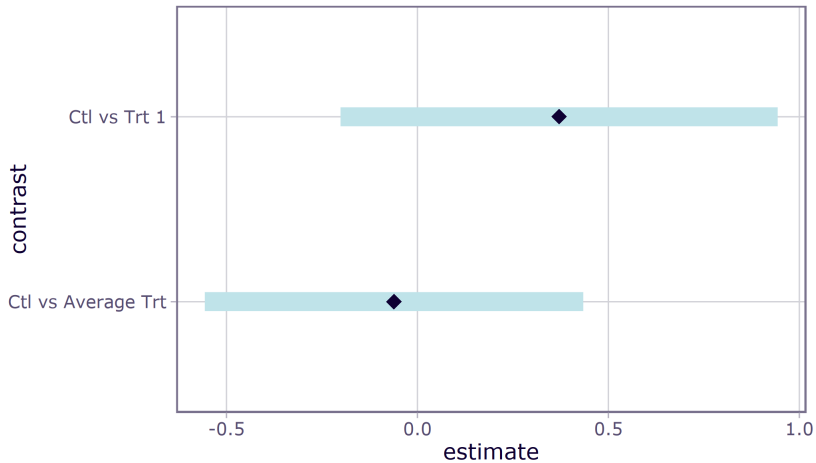
```
# pg_model <- aov(weight ~ group, data = PlantGrowth)
# pg_emm <- emmeans(pg_model, ~ group)
custom_list <- list(
  "Ctl vs Average Trt" = c(1, -0.5, -0.5),
  "Ctl vs Trt 1" = c(1, -1, 0)
)
contrast(pg_emm, method = custom_list)
```

contrast	estimate	SE	df	t.ratio	p.value
Ctl vs Average Trt	-0.0615	0.241	27	-0.255	0.8009
Ctl vs Trt 1	0.3710	0.279	27	1.331	0.1944

Contrasts in R II

```
pg_cont <- contrast(pg_emm, method = custom_list)
plot(pg_cont)  # level option to change 1-alpha
```

Contrasts in R III



Multiple Comparisons

Basic Error Types

- A test of a single hypothesis is characterized by two error rates: the Type I error rate (α) and the Type II error rate (β).

The Need for Multiple Comparisons

- When picking contrasts, it is important to consider the issue of **multiple comparisons**.
- Idea: we pick a α value to test at (e.g., $\alpha = 0.05$). That means, for a single test, there is only a 5% chance (1/20) that we will “incorrectly reject” a true null. **What if we conduct 15 tests?**

Types of Error Rates

When conducting K hypotheses, the situation becomes more complicated, and we must define which type of error rate we to control.

- **Per Comparison Error Rate (PCER)**: The probability of a Type I error for a single, specific hypothesis test (standard α).
- **Familywise Error Rate (FWER)**: The probability of making at least one Type I error among K tests, assuming *all* null hypotheses are true.
- **Strong Familywise Error Rate (Strong FWER)**: The probability of making at least one Type I error among true nulls, *regardless* of how many null hypotheses are true or false.

Types of Error Rates II

- **False Discovery Rate (FDR)**: The expected proportion of false discoveries (false rejections) among all rejected null hypotheses:

$$\text{FDR} = E \left[\frac{\text{False Rejections}}{\text{Total Rejections}} \right]$$

General Methods: Introduction

- We will start by introducing some general methods that can be applied *regardless* of the model assumptions or type of hypothesis being adjusted.
- The two most general are the **Bonferroni** and **Holm** adjustment methods, that require no assumptions.
- These can be easily implemented using the `p.adjust` method in R:

```
p.adjust(  
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),  
  method = 'holm'  
)  
  
[1] 0.136 0.006 0.168 0.280 0.168 0.100
```

General Methods: Bonferroni Adjustment

- **Description:** A simple, conservative adjustment based on the Bonferroni inequality ($P(\bigcup E_i) \leq \sum P(E_i)$). Makes no distributional assumptions and works for any set of K tests.
- **Error Controlled:** Controls the **Strong Familywise Error Rate (Strong FWER)** at level α .
- **How-To Implement:**
 - *Adjust Significance Level:* Reject $H_{0,i}$ if $p_i \leq \alpha_{\text{adj}} = \frac{\alpha}{K}$.
 - *Adjust p-values:* Multiply raw p -values by K :

$$p_{i,\text{adj}} = \min(1, K \times p_i)$$

- **When to Use:** Best for a small, pre-planned set of $K \leq 6$ contrasts. Becomes overly conservative as K increases.

General Methods: Bonferroni Adjustment II

```
p.adjust(  
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),  
  method = 'bonferroni'  
)
```

```
[1] 0.204 0.006 0.336 1.000 0.360 0.120
```

General Methods: Holm's Step-Down Method

- **Description:** A sequential step-down modification of Bonferroni. It is strictly and uniformly more powerful than standard Bonferroni while making zero distributional or independence assumptions.
- **Error Controlled:** Controls the **Strong Familywise Error Rate (Strong FWER)** at level α .
- **How-To Implement::**
 1. Sort raw p -values from smallest to largest:
 $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(K)}.$
 2. Compare each $p_{(i)}$ sequentially to the adaptive threshold:

$$\text{Threshold}_i = \frac{\alpha}{K - i + 1}$$

General Methods: Holm's Step-Down Method II

3. Start at $i = 1$. As long as $p_{(i)} \leq \frac{\alpha}{K-i+1}$, reject $H_{(i)}$ and proceed to $i + 1$.
 4. **Stop Rule:** The moment $p_{(i)} > \frac{\alpha}{K-i+1}$, **stop testing** and retain all remaining null hypotheses $H_{(i)}, \dots, H_{(K)}$.
- Alternatively, we can adjust the p-values using the formula:

$$p_{\text{adj},i} = \max_{j \leq i} \left\{ \min \left\{ (K - j + 1)p_{(j)}, 1 \right\} \right\}$$

```
p.adjust(  
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),  
  method = 'holm'  
)
```

```
[1] 0.136 0.006 0.168 0.280 0.168 0.100
```

General Methods: Benjamini-Hochberg (FDR)

- **Description:** A sequential step-up procedure designed to maximize power when testing hundreds or thousands of hypotheses (e.g., genomics, large screening studies).
- **Error Controlled:** Controls the **False Discovery Rate (FDR)** at target level q (e.g., $q = 0.05$).
- **How-To Implement:**
 1. Sort raw p -values from smallest to largest:
$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(K)}.$$
 2. Calculate the FDR critical boundary for each rank i :

$$\text{Critical Value}_i = \left(\frac{i}{K} \right) q$$

3. Find the *largest* rank k such that $p_{(k)} \leq \left(\frac{k}{K} \right) q$.

General Methods: Benjamini-Hochberg (FDR) II

4. **Decision Rule:** Reject $H_{(1)}, H_{(2)}, \dots, H_{(k)}$. (All hypotheses ranked up to k are declared significant, even if individual $p_{(i)} > \text{Critical Value}_i$ for $i < k$).

```
p.adjust(  
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),  
  method = 'BH'  # or 'fdr', both give the same result  
)
```

```
[1] 0.068 0.006 0.072 0.280 0.072 0.060
```

ANOVA specific methods

- If we want to use multiple-testing approaches for ANOVA and contrasts, any of the above are perfectly valid.
- However, we can often use our model assumptions to get more powerful methods specific to ANOVA and our linear models.
- We will next describe a few key methods in this scenario.

ANOVA Methods: Tukey's HSD (All Pairwise)

- **Description:** Tailored specifically for comparing all possible pairwise means ($\binom{g}{2}$ comparisons). Uses the Studentized Range distribution (Q) rather than the standard t -distribution.
- **Error Controlled:** Controls the **Strong Familywise Error Rate (Strong FWER)** at level α specifically across the family of all pairwise comparisons.
- **How-To Implement:**
 - Compute the Honest Significant Difference (HSD) threshold:

$$\text{HSD} = q_{1-\alpha}(g, N - g) \sqrt{\frac{MS_E}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

- Declare two group means significantly different if $|\bar{y}_{i\bullet} - \bar{y}_{j\bullet}| \geq \text{HSD}$.

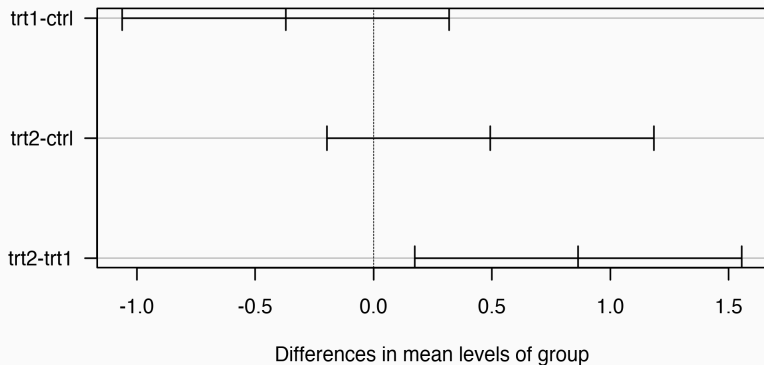
ANOVA Methods: Tukey's HSD (All Pairwise) II

- **Unequal Sample Sizes:** Modern software automatically incorporates the **Tukey-Cramer** modification above, adjusting SE for unequal n_i and n_j .

```
# pg_model <- aov(weight ~ group, data = PlantGrowth)
# pg_emm <- emmeans(pg_model, ~ group)
tukeyRes <- TukeyHSD(pg_model) # Compute Tukey Adjusted Intervals
plot(tukeyRes, las = 1) # Plot Tukey Adjusted Intervals
```

ANOVA Methods: Tukey's HSD (All Pairwise) III

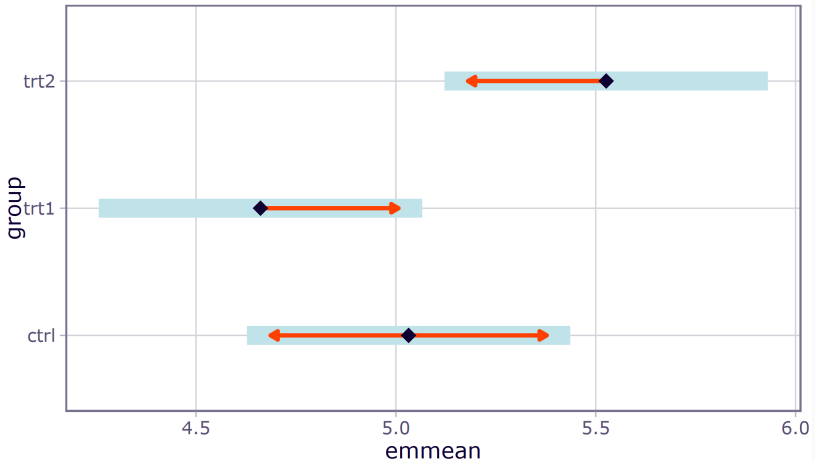
95% family-wise confidence level



ANOVA Methods: Tukey's HSD (All Pairwise) IV

```
# Default adjust = 'tukey'.  
# Group means differ significantly if arrows don't overlap.  
plot(pg_emm, comparisons = TRUE)
```

ANOVA Methods: Tukey's HSD (All Pairwise) V



ANOVA Methods: Dunnett's Test (Control vs Treatments)

- **Description:** Designed specifically for experiments with one baseline control group and $g - 1$ treatment groups. Uses a multivariate t -distribution that accounts for the shared correlation between comparisons involving the same control group.
- **Error Controlled:** Controls the Strong Familywise Error Rate (Strong FWER) at level α across the $g - 1$ treatment-vs-control comparisons.
- **How-To Implement:**
 - Compute the Dunnett threshold for group i vs control group c :

$$D = d_{1-\alpha}(g - 1, N - g) \sqrt{MS_E \left(\frac{1}{n_i} + \frac{1}{n_c} \right)}$$

ANOVA Methods: Dunnett's Test (Control vs Treatments) II

- Reject $H_0 : \mu_i = \mu_c$ if $|\bar{y}_{i\bullet} - \bar{y}_{c\bullet}| \geq D$.
- **Advantage:** Substantially more powerful than Tukey or Bonferroni when you only care about comparisons against a control group.

ANOVA Methods: Dunnett's Test (Control vs Treatments) III

```
# pg_model <- aov(weight ~ group, data = PlantGrowth)
# pg_emm <- emmeans(pg_model, ~ group)

# Compute Dunnett-adjusted contrasts against baseline 'ctrl'
dunnettRes <- contrast(
  pg_emm,
  method = "trt.vs.ctrl",
  ref = "ctrl"
)
summary(dunnettRes)
```

contrast	estimate	SE	df	t.ratio	p.value
trt1 - ctrl	-0.371	0.279	27	-1.331	0.3296
trt2 - ctrl	0.494	0.279	27	1.772	0.1582

P value adjustment: dunnettx method for 2 tests

ANOVA Methods: Scheffé's Method (Data Snooping)

- **Description:** Based on the omnibus ANOVA F -distribution. It is the most conservative post-hoc method, protecting against false positives across *all possible* linear contrasts.
- **Error Controlled:** Controls the **Strong Familywise Error Rate (Strong FWER)** at level α across the *infinite set* of all possible linear contrasts.
- **How-To Implement:**
 - For any contrast $\hat{W} = \sum w_i \bar{y}_{i\bullet}$, compute the Scheffé Critical Value (SSD):

$$\text{SSD} = \sqrt{(g-1) F_{1-\alpha, g-1, N-g}} \times \text{SE}(\hat{W})$$

$$\text{where } \text{SE}(\hat{W}) = \sqrt{MS_E \sum \frac{w_i^2}{n_i}}.$$

ANOVA Methods: Scheffé's Method (Data Snooping) II

- Reject $H_0 : W = 0$ if $|\hat{W}| \geq \text{SSD}$.
- **When to Use:** Use when testing custom contrasts constructed *after* inspecting the data ("data snooping") or testing a large number of non-pairwise complex contrasts.

ANOVA Methods: Scheffé's Method (Data Snooping) III

```
# Define custom complex contrasts  
# Example: Control vs Average of Treatments, and Trt1 vs Trt2  
custom_contrasts <- list(  
  "Ctrl vs Avg(Trt1, Trt2)" = c(1, -0.5, -0.5),  
  "Trt1 vs Trt2"            = c(0, 1, -1)  
)
```

```
# Compute Scheffe-adjusted contrasts  
scheffeRes <- contrast(pg_emm, method = custom_contrasts,  
                       adjust = "scheffe")  
summary(scheffeRes)
```

contrast	estimate	SE	df	t.ratio
Ctrl vs Avg(Trt1, Trt2)	-0.0615	0.241	27	-0.255
Trt1 vs Trt2	-0.8650	0.279	27	-3.103
p.value				
0.9681				

Simultaneous Confidence Intervals

- **Definition:** A set of intervals $[L_1, U_1], \dots, [L_K, U_K]$ are $(1 - \alpha)$ **simultaneous confidence intervals** if the probability that *all* K intervals cover their true parameter values $\theta_1, \dots, \theta_K$ simultaneously is at least $1 - \alpha$:

$$P \left(\bigcap_{k=1}^K \{ \theta_k \in [L_k, U_k] \} \right) \geq 1 - \alpha$$

- **General Structure:** For linear contrasts $\theta_k = \sum w_i \mu_i$, simultaneous intervals take the standard form:

$$\hat{\theta}_k \pm M_\alpha \times \text{SE}(\hat{\theta}_k)$$

where M_α is an adjusted critical multiplier derived from the multiple testing procedure:

Simultaneous Confidence Intervals II

- **Bonferroni:** $M_\alpha = t_{1-\frac{\alpha}{2K}, N-g}$
- **Tukey's HSD:** $M_\alpha = \frac{1}{\sqrt{2}} q_{1-\alpha, g, N-g}$
- **Scheffé:** $M_\alpha = \sqrt{(g-1) F_{1-\alpha, g-1, N-g}}$

Duality: Simultaneous CIs and Strong FWER

- **The Duality Principle:** Hypothesis tests and confidence intervals are inverted versions of the same mathematical statement:

$$\text{Reject } H_{0,k} : \theta_k = 0 \text{ at level } \alpha \iff 0 \notin \text{CI}_k^{(1-\alpha)}$$

- **Why Strong FWER is Required:**
 - *Weak FWER* only guarantees protection when *all* null hypotheses are true ($\theta_1 = \theta_2 = \dots = \theta_K = 0$).
 - In practice, some contrasts will be zero and others will be non-zero. Confidence intervals must cover the true parameters **regardless of which or how many parameters are non-zero.**

Duality: Simultaneous CIs and Strong FWER II

- **Strong FWER** controls the Type I error rate under *any* combination of true and false null hypotheses. Inverting a test procedure that controls Strong FWER guarantees $(1 - \alpha)$ simultaneous coverage across all parameters.
- **Practical Takeaway:** When presenting a plot or table of Tukey, Bonferroni, or Scheffé simultaneous confidence intervals, any interval that excludes 0 corresponds *exactly* to a contrast that is statistically significant under that method's Strong FWER-adjusted test at level α .

Connecting Theory to R: Extracting M_α and CIs

- Modern R packages like `emmeans` apply the critical multiplier M_α automatically when you request confidence intervals.
- **Duality in Output:** Running `summary()` displays test statistics (p -values), while `confint()` displays simultaneous confidence intervals derived using the exact same M_α .

Connecting Theory to R: Extracting M_α and CIs II

```
library(emmeans)
# 1. Fit model and get pairwise contrasts
pg_model <- aov(weight ~ group, data = PlantGrowth)
pg_emm    <- emmeans(pg_model, ~ group)
pg_pairs  <- pairs(pg_emm, adjust = "tukey")
```

```
# 2. View Hypothesis Testing (p-values)
summary(pg_pairs)
```

contrast	estimate	SE	df	t.ratio	p.value
ctrl - trt1	0.371	0.279	27	1.331	0.3909
ctrl - trt2	-0.494	0.279	27	-1.772	0.1980
trt1 - trt2	-0.865	0.279	27	-3.103	0.0120

P value adjustment: tukey method for comparing a family of 3 est

```
# 3. View Simultaneous Confidence Intervals (Estimate +/- M_alpha
```

Summary: Selecting an Adjustment Method

- No single adjustment method is universally “best”—the choice depends entirely on the structure of your research question and the family of tests.

Method	Target Error Rate	Family Scope	Best Application
Bonferroni	Strong FWER	General (K tests)	Small set of pre-planned contrasts ($K \leq 6$)
Holm	Strong FWER	General (K tests)	Step-down alternative to Bonferroni (more power)
BH / FDR	FDR	General (K tests)	High-throughput screening / large K
Tukey / HSD	Strong FWER	All Pairwise ($\binom{g}{2}$)	Comparing all treatment group pairs
Dunnett	Strong FWER	Control vs All ($g - 1$)	Comparing all treatments against one control
Scheffé	Strong FWER	Infinite Contrasts	Data snooping / Complex post-hoc contrasts

Applied Example: Plant Growth

Example 1: Data Setup and the F-Test

- We will use R's built-in `PlantGrowth` dataset, measuring dried plant yields under a control and two different treatment conditions.
- As discussed, a Completely Randomized Design (CRD) is the simplest design for comparing treatments. We have $g = 3$ groups and $N = 30$ total plants.

Example 1: Data Setup and the F-Test II

```
# Load data and check the structure
data(PlantGrowth)
str(PlantGrowth)

'data.frame': 30 obs. of 2 variables:
 $ weight: num 4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.5..
 $ group : Factor w/ 3 levels "ctrl","trt1",...: 1 1 1..

# Ensure the treatment is a factor (categorical)
class(PlantGrowth$group)

[1] "factor"
```

Example 1: Data Setup and the F-Test III

- We first test the null hypothesis that all group means are equal using an omnibus ANOVA F-test.

```
# Run the ANOVA model
pg_model <- aov(weight ~ group, data = PlantGrowth)
```

```
# View the ANOVA table
summary(pg_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	3.766	1.8832	4.846	0.0159 *
Residuals	27	10.492	0.3886		

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example 1: Data Setup and the F-Test IV

- **Interpretation:** With a p -value of 0.0159, we reject the null hypothesis. There is a significant difference in plant yields between at least some of the groups. But which ones?

Example 2: Formulating Custom Contrasts

- We know from our model that the single mean model is inadequate, so we want to estimate the separate group means.
- Let's test a specific contrast: Is the control group significantly different from the average of the two experimental treatments?

Example 2: Formulating Custom Contrasts II

```
library(emmeans)

# Get estimated marginal means
pg_emm <- emmeans(pg_model, ~ group)

# Define contrast: Control (1) vs Average of Trt1 (2) and Trt2 (3)
# Weights must sum to zero
custom_list <- list("Control vs Average Trt" = c(1, -0.5, -0.5))

# Test the contrast
contrast(pg_emm, method = custom_list)
```

contrast	estimate	SE	df	t.ratio
Control vs Average Trt	-0.0615	0.241	27	-0.255
p.value	0.8009			

Example 2: Formulating Custom Contrasts III

- **Interpretation:** The p -value is 0.2644. We do not have sufficient evidence to say the control is different from the *average* of the two treatments.

Example 3: Multiple Comparisons

- What if we want to compare all individual pairs of treatments?
- We must adjust for multiple comparisons to control the familywise error rate. Tukey's HSD is the preferred method for all pairwise comparisons.

```
# Run all pairwise comparisons with Tukey adjustment  
pg_pairwise <- contrast(pg_emm, method = "pairwise")
```

```
summary(pg_pairwise)
```

contrast	estimate	SE	df	t.ratio	p.value
ctrl - trt1	0.371	0.279	27	1.331	0.3909
ctrl - trt2	-0.494	0.279	27	-1.772	0.1980
trt1 - trt2	-0.865	0.279	27	-3.103	0.0120

P value adjustment: tukey method for comparing a family of 3 estimates

Example 3: Multiple Comparisons II

- **Interpretation:** After applying the Tukey penalty, only the difference between trt1 and trt2 is statistically significant ($p = 0.0120$). The control is not significantly different from either individual treatment.

Other Common Scenarios (See Textbook)

- While Holm and Tukey methods cover many of basic the basic contrasts we may wish to test, there are specialized methods for other common scenarios.
- **Further Reading:** We will not cover the mathematical derivations of these methods in class, but you can find their details, applications, and statistical tables in Oehlert (2000). For instance, comparing all to treatments to the “best” treatment (section 5.5.2), not known in advance.

Testing Model Assumptions

Core Assumptions of the Errors

- The validity of ANOVA results, including contrasts and some forms of multiple testing, depends on our model assumptions.
- For the one-way ANOVA CRD model: $y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$.
- Our main assumptions for this model include:
 - All ϵ_{ij} are independent
 - The variances of ϵ_{ij} 's are all the same.
 - The ϵ_{ij} 's are normally distributed.

Core Assumptions of the Errors II

- Listing the assumptions in decreasing order of importance:
 1. **Independence** (most important).
 2. **Constant variance** (next most important).
 3. **Normal distribution** (least important).

Diagnosing Violations

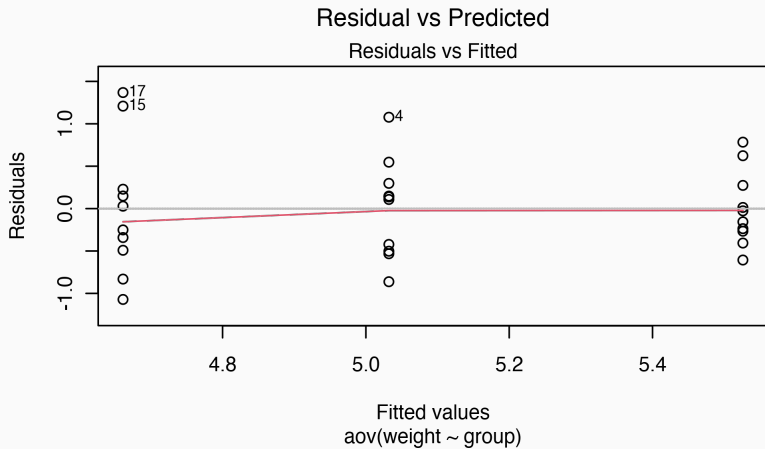
- Essentially all methods for diagnosing violation of assumptions are based on study of the **residuals**.
- Conceptually: $r_{ij} = \text{observed}_{ij} - \text{predicted}_{ij}$. Intuitively, if your model is appropriate, r_{ij} should resemble ϵ_{ij} .
- **Tests vs. Plots:** There are tests for normality and equal variances, but they have limitations (small sample sizes lack power; large sample sizes detect trivial problems).
- Therefore, formal tests are *supplemental*, while plots are *principal*.

R Code: Principal Plots for Diagnostics

- We can generate the principal diagnostic plots (Residual vs Predicted and Normal QQ Plot) directly from our ANOVA model object in R.

```
# Using the pg_model (PlantGrowth) from earlier  
# 1. Residual vs Predicted (Fitted) plot for constant variance  
plot(pg_model, which = 1, main = "Residual vs Predicted")
```

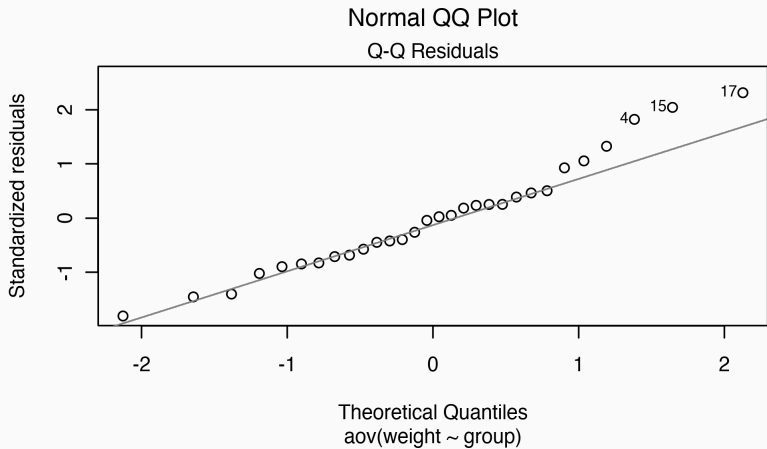
R Code: Principal Plots for Diagnostics II



R Code: Principal Plots for Diagnostics III

```
# 2. Normal probability (QQ) plot for normality  
plot(pg_model, which = 2, main = "Normal QQ Plot")
```

R Code: Principal Plots for Diagnostics IV



R Code: Principal Plots for Diagnostics V

- **Supplemental Tests:** If you need formal statistical tests, you can use the Shapiro-Wilk test for normality and Levene's test for equal variances.

R Code: Principal Plots for Diagnostics VI

```
# Supplemental Test for Normality on residuals  
# (Reject Normality assumption if small)  
shapiro.test(resid(pg_model))
```

Shapiro-Wilk normality test

```
data: resid(pg_model)  
W = 0.96607, p-value = 0.4379
```

```
# Supplemental Test for Equal Variances (requires 'car' package)  
library(car)  
leveneTest(weight ~ group, data = PlantGrowth)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	1.1192	0.3412
	27		

Checking Specific Assumptions

- **Independence:** Independence of the ϵ_{ij} 's cannot be checked by considering residuals.

Checking Specific Assumptions II

- **Variance and Normality:** Heteroskedasticity (changing variances) and non-normality usually come hand-in-hand. Both can be checked by formal tests (supplemental), or by plots (primary approach). Key plots include:
 - Normal QQ-plots
 - Residuals vs predicted plot
 - Residuals in time order plot

Remedies: Data Transformations

- The principle remedial tool is re-expression (transformation) of the response.
- If two transformations $\tilde{y}_1 = f_1(y)$ and $\tilde{y}_2 = f_2(y)$ have a linear relationship ($\tilde{y}_2 = \frac{\tilde{y}_1 - a}{b}$), they are completely equivalent for correcting assumptions.
- For log transformations, the transformations $\tilde{y}_1 = \log_e(y)$ and $\tilde{y}_2 = \log_{10}(y)$ are equivalent transformations because $\tilde{y}_1 = \ln(10)\tilde{y}_2$.

Remedies: Data Transformations II

- Another common transformation includes the Box-Cox power family of transformations for a positive response variable is defined as:

$$\tilde{y} = \begin{cases} (y^p - 1)/p, & p \neq 0 \\ \ln(y), & p = 0 \end{cases}$$

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

Wood T, Bormann F (1974). "The effects of an artificial acid mist upon the growth of *Betula alleghaniensis* Britt." *Environmental Pollution* (1970), **7**(4), 259–268.

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References and Acknowledgements II

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